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MATHS MASTER _____

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CANDIDATE NUMBER

2026 Trial Examination

Form VI Mathematics Extension 2

Wednesday 12th August, 2026

8:40am

General Instructions

- Reading time — 10 minutes
- Working time — 3 hours
- Attempt all questions.
- Write using black pen.
- Calculators approved by NESA may be used.
- A loose reference sheet is provided separate to this paper.

Sixteen Questions — 100 Marks

Section I (10 marks) Questions 1 – 10

- This section is multiple choice. Each question is worth 1 mark.
- Record your answers on the provided answer sheet.

Section II (90 marks) Questions 11 – 16

- Each question is worth 15 marks.
- Relevant mathematical reasoning and calculations are required.
- Start each question in a new booklet.

Collection

- Your name and master should only be written on this page.
- Write your candidate number on this page, on each booklet and on the multiple-choice answer sheet.
- If you use multiple booklets for a question, place them inside the first booklet for the question.
- Arrange your solutions in order.

Checklist

- Reference sheet
- Multiple-choice answer sheet
- 6 booklets per boy
- Candidature: 70 pupils

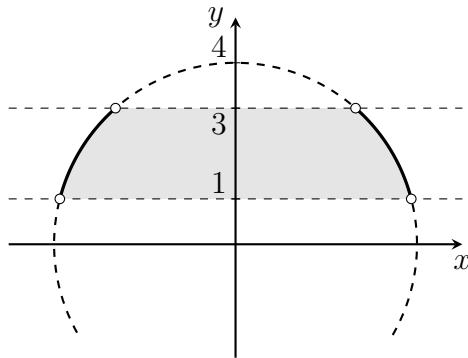
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Section I

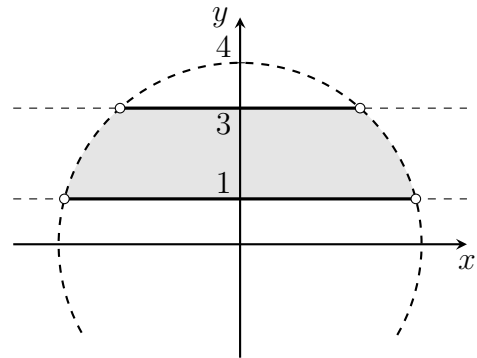
1. Let R be the region in the complex plane defined by $1 \leq \operatorname{Im}(z) \leq 3$ and $|z| < 4$. Which diagram best represents the region R ?

1

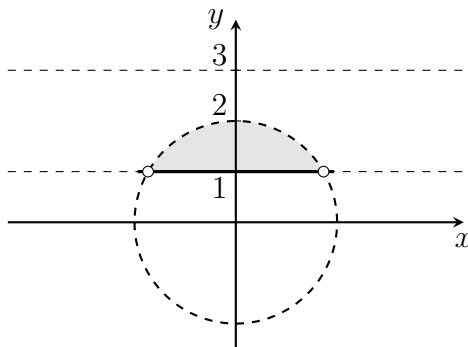
(A)



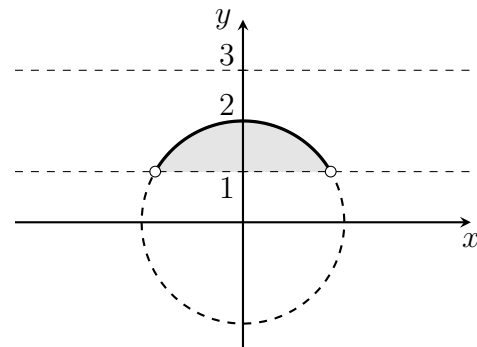
(B)



(C)



(D)



2. A driving manual contains the following statement:

1

‘If someone is bad at driving, then they will never miss an exit.’

Which of the following is logically equivalent to this statement?

- (A) If someone is good at driving, then they will never miss an exit.
- (B) If someone is not bad at driving, then they will miss an exit.
- (C) If someone misses an exit, then they are bad at driving.
- (D) If someone misses an exit, then they are not bad at driving.

3. Which of the following is a correct expression for $\int \frac{dx}{\sqrt{5-4x-x^2}}$? 1

(A) $\sin^{-1}\left(\frac{x-2}{3}\right) + c$

(B) $\sin^{-1}\left(\frac{x+2}{3}\right) + c$

(C) $\sin^{-1}\left(\frac{x-2}{9}\right) + c$

(D) $\sin^{-1}\left(\frac{x+2}{9}\right) + c$

4. A complex number $z = x + iy$ is defined such that $\operatorname{Re}\left(\frac{1}{z}\right) = \frac{1}{2}$. 1

Which of the following circles in the complex plane must z always lie on?

(A) $(x-1)^2 + y^2 = 1$

(B) $x^2 + (y-1)^2 = 1$

(C) $(x+1)^2 + y^2 = 1$

(D) $x^2 + (y+1)^2 = 1$

5. The vectors $\underline{p}$ and $\underline{q}$ are given by $\underline{p} = \begin{pmatrix} \sin \theta \\ \cos \theta \\ \tan \theta \end{pmatrix}$ and $\underline{q} = \begin{pmatrix} \operatorname{cosec} \theta \\ \sec \theta \\ \cot \theta \end{pmatrix}$, where $0 \leq \theta \leq 2\pi$. 1

For how many values of θ is $\operatorname{proj}_{\underline{p}} \underline{q}$ a unit vector?

(A) 0

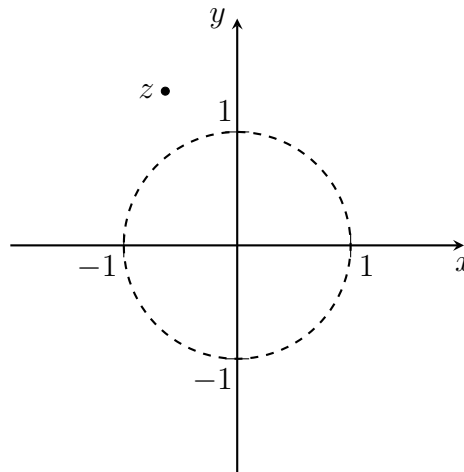
(B) 1

(C) 2

(D) 4

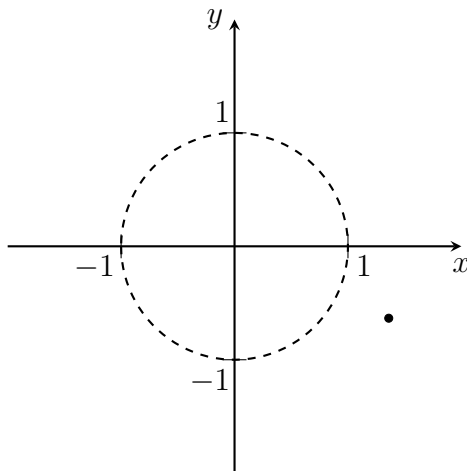
6. The diagram below shows the complex number $z = re^{i\theta}$ on the Argand diagram.

1

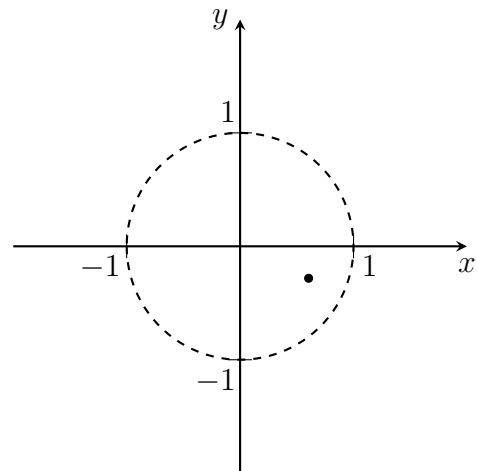


Which of the following diagrams best shows the position of $\frac{ir^2}{z}$?

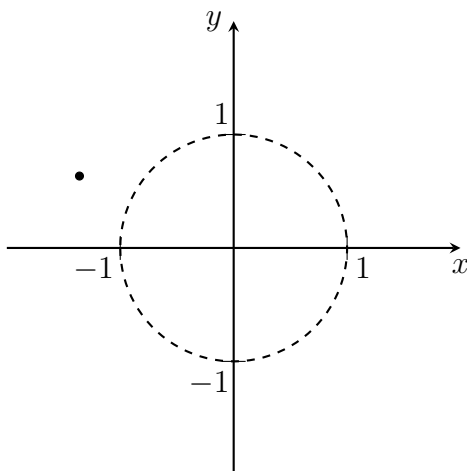
(A)



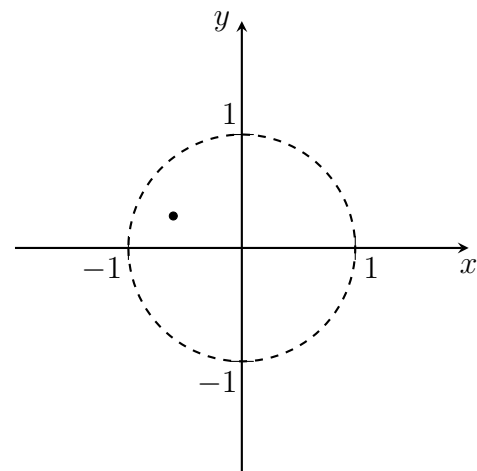
(B)



(C)



(D)



7. Consider the statement:

1

$$\forall p, q \in \mathbb{Z}^+, \exists m, n \in \mathbb{Z}^+ \text{ such that } \sqrt{p+q} = \sqrt{m} + \sqrt{n}.$$

Which of the following could be used as a counterexample to disprove this statement?

- (A) $p = 2, q = -2$
- (B) $p = 6, q = 3$
- (C) $p = 10, q = 8$
- (D) $p = 2, q = 4$
8. It is known that the vector $\underline{r} = \underline{i} - \underline{j} + 2\underline{k}$ is perpendicular to $a\underline{i} + b\underline{j} + c\underline{k}$, where $a, b, c \neq 0$. Which of the following vectors is NOT perpendicular to $\underline{r}$?
- (A) $(a+c)\underline{i} + (b+c)\underline{j} + c\underline{k}$
- (B) $b\underline{i} + a\underline{j} - c\underline{k}$
- (C) $b\underline{i} + a\underline{j} + (a-b+c)\underline{k}$
- (D) $(2c-a)\underline{i} + (b-2c)\underline{j} + (a+b-c)\underline{k}$

1

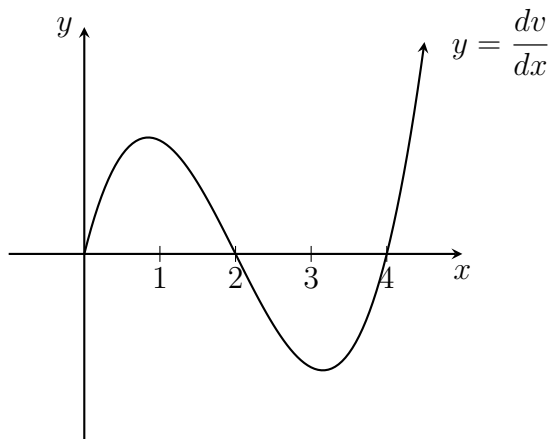
9. The complex number z is defined such that $-\frac{\pi}{4} \leq \text{Arg}(z-i) \leq \frac{\pi}{4}$.

1

Which of the following describes the set of values that $\text{Arg}(z^2+1)$ can take?

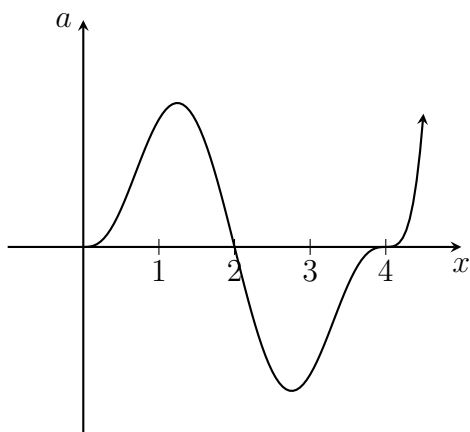
- (A) $-\frac{\pi}{2} < \text{Arg}(z^2+1) < \frac{\pi}{2}$
- (B) $-\frac{\pi}{2} < \text{Arg}(z^2+1) < \frac{3\pi}{4}$
- (C) $-\frac{3\pi}{4} < \text{Arg}(z^2+1) < \frac{\pi}{4}$
- (D) $-\frac{\pi}{4} < \text{Arg}(z^2+1) < \pi$

10. The graph below shows the derivative of the velocity v of a particle as a function of its displacement x , where $x \geq 0$.

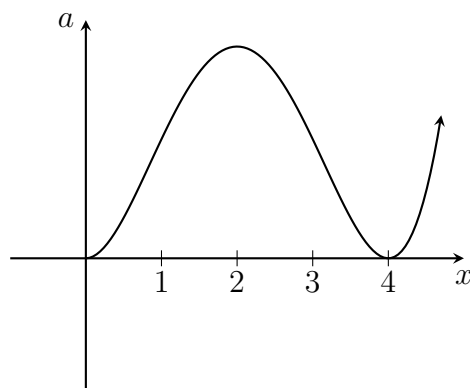


It is known that if the particle is at $x = 0$, it will be stationary and will not move. Which of the following graphs best shows the acceleration of the particle as a function of its displacement?

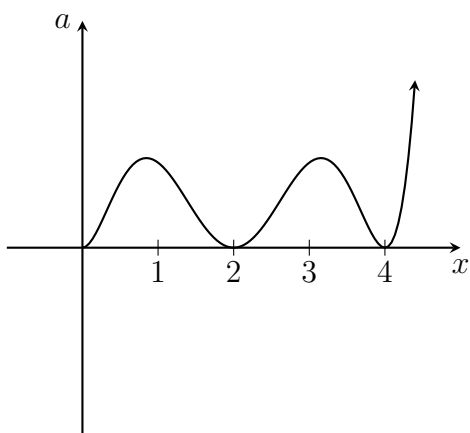
(A)



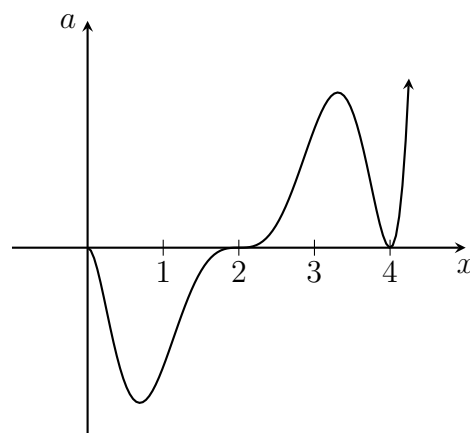
(B)



(C)



(D)



End of Section I

The paper continues in the next section

Section II

This section consists of long-answer questions.

Marks may be awarded for reasoning and calculations.

Marks may be lost for poor setting out or poor logic.

Start each question in a new booklet.

QUESTION ELEVEN (15 marks) Start a new answer booklet.

(a) Solve $z^2 - 4z + 8 = 0$.

2

(b) The complex numbers w and z are given by $w = 8e^{\frac{3i\pi}{4}}$ and $z = 2e^{\frac{i\pi}{6}}$.

2

Find the modulus and principal argument of $\frac{w}{z}$.

(c) Find the following:

(i) $\int (\tan^2 x + 1) \sec^2 x \, dx$

2

(ii) $\int \frac{x^2}{4 + x^2} \, dx$

2

(iii) $\int x \cos x \, dx$

2

(d) Two lines are given by $\underline{r}_1 = \begin{pmatrix} 1 \\ -2 \\ 14 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix}$ and $\underline{r}_2 = \begin{pmatrix} -6 \\ 3 \\ -1 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$,

where $\lambda, \mu \in \mathbb{R}$.

(i) Show that the lines intersect, and find their point of intersection.

3

(ii) Determine the acute angle between the lines, giving your answer correct to the nearest degree.

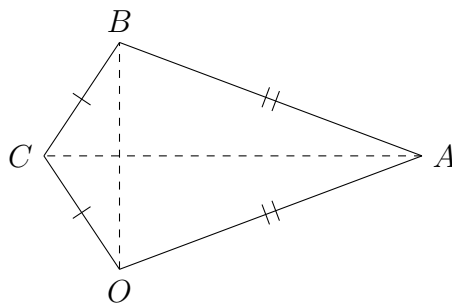
2

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QUESTION TWELVE (15 marks) Start a new answer booklet.

- (a) (i) Show that if n is an integer, then $n^2 + 7n + 6$ is even. **2**
- (ii) Hence determine, with justification, the number of integer solutions to the equation $(n^2 + 7n + 6)^7 - 16n^5 = 3$. **1**
- (b) (i) Find the values of A , B , and C such that **2**
- $$\frac{3x^2 - 6x + 13}{(x - 5)(x^2 + 4)} = \frac{A}{x - 5} + \frac{Bx + C}{x^2 + 4}.$$
- (ii) Hence find $\int \frac{3x^2 - 6x + 13}{(x - 5)(x^2 + 4)} dx$. **1**
- (c) An object moving in simple harmonic motion has displacement x m and velocity v ms⁻¹, and its motion satisfies $v^2 = 27 - 18x - 9x^2$.
- (i) Find the centre of motion and the amplitude. **2**
- (ii) Determine the period and hence find the distance that the particle will travel in 30π seconds. **2**
- (iii) If the object has a mass of 5 kg, find the maximum force that acts on the object throughout its motion. **1**
- (d) The equation $x^3 + bx^2 + cx + 3b = 0$, where b and c are real, has a complex root $b + ic$. Using the relationship between the roots and the coefficients:
- (i) show that the equation has a real root of $-3b$, **1**
- (ii) find the possible values of b and c . **3**

The paper continues on the next page

QUESTION THIRTEEN (15 marks) Start a new answer booklet.(a) Consider the statement: ‘If n is a multiple of 6, then $(1 - i\sqrt{3})^n$ is real’.(i) Prove that the statement is true. 2(ii) State the converse and determine if it is true, justifying your answer. 2(b) Use integration by parts to find $\int e^x \cos 2x \, dx$. 3(c) The diagram below shows the quadrilateral $OABC$, where $OA = BA$ and $OC = BC$.Let $\overrightarrow{OA} = \underline{a}$, $\overrightarrow{OB} = \underline{b}$ and $\overrightarrow{OC} = \underline{c}$.(i) Use the fact that $|\underline{a}| = |\underline{b} - \underline{a}|$ to show that $\underline{b} \cdot \underline{b} = 2\underline{a} \cdot \underline{b}$. 1(ii) Use vector methods to show that $OB \perp AC$. 2(d) (i) By considering the expansion of $(\cos \theta + i \sin \theta)^4$, show that 3

$$\tan 4\theta = \frac{4 \tan \theta - 4 \tan^3 \theta}{1 - 6 \tan^2 \theta + \tan^4 \theta}.$$

(ii) Hence find the roots of the polynomial $x^4 - 4x^3 - 6x^2 + 4x + 1 = 0$. 2

The paper continues on the next page

QUESTION FOURTEEN (15 marks) Start a new answer booklet.

- (a) On a distant planet, an astronaut determines that a particular object of mass m kg experiences a resistive force of mkv^2 when it moves through the atmosphere, where v is the velocity of the object and k is a positive constant. The magnitude of acceleration due to gravity on the planet is $G \text{ ms}^{-2}$.

When falling, the equation of motion of the particle is $\ddot{x} = -G + kv^2$, and the astronaut determines that the object has a terminal velocity of 40 ms^{-1} in the negative direction.

(i) Show that $\sqrt{\frac{G}{k}} = 40$.

1

- (ii) The object is projected upwards with a velocity of 40 ms^{-1} . When it is moving upwards, its equation of motion is $\ddot{x} = -G - kv^2$.

(α) Show that $v = 40 \tan\left(\frac{\pi}{4} - 40kt\right)$.

2

- (β) The astronaut observes that it takes 4 seconds for it to reach its maximum height. Calculate the exact value of G .

2

- (b) Two complex numbers z_1 and z_2 lie on the unit circle, and are given by $z_1 = e^{i\alpha}$ and $z_2 = e^{i\beta}$ where $0 \leq \alpha \leq \pi$ and $0 \leq \beta \leq \pi$.

(i) Show that $e^{i\theta} + e^{-i\theta} = 2 \cos \theta$.

1

(ii) The complex number z_3 is defined by $z_3 = \frac{1}{z_1 + z_2}$.

3

For certain values of α and β , it is found that z_3 has modulus $\frac{1}{\sqrt{3}}$.

Using the fact that

$$e^{i\alpha} + e^{i\beta} = e^{i(\alpha+\beta)/2} (e^{i(\alpha-\beta)/2} + e^{-i(\alpha-\beta)/2}),$$

or otherwise, find the possible values of $\alpha - \beta$.

- (c) (i) Use mathematical induction to show that $x + 1$ is a factor of $x^n + 1$ for all **odd** positive integers n .

2

- (ii) It is known that for some $n, k, m \in \mathbb{Z}^+$, $n = m \times 2^k$.

2

Use proof by contraposition to show that if n is not a power of 2, then m has an odd factor larger than 1.

- (iii) Deduce that if $2^n + 1$ is prime, then n is a power of 2.

2

The paper continues on the next page

QUESTION FIFTEEN (15 marks) Start a new answer booklet.

(a) Let $I_n = \int_0^1 (1 - \sqrt{x})^n dx$ for integers $n \geq 0$.

(i) Use integration by parts to show that $I_n = \frac{n}{n+2} I_{n-1}$ for $n \geq 1$. **3**

(ii) Hence show that $I_n = \frac{2}{(n+1)(n+2)}$. **1**

(iii) By considering the partial fractions decomposition of I_n , or otherwise, deduce that $I_0 + I_1 + I_2 + \cdots + I_n < 2$. **2**

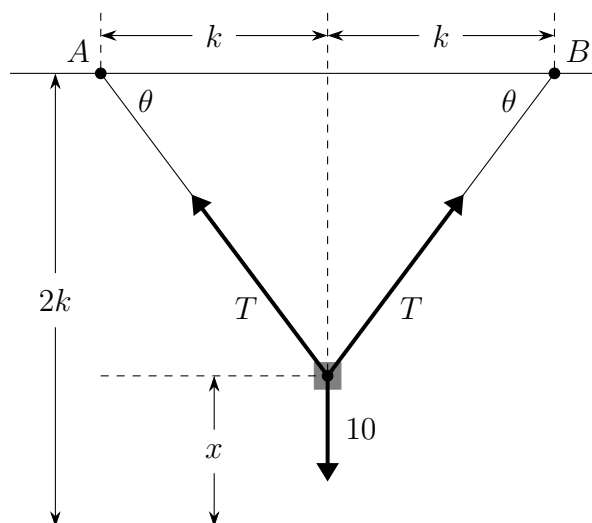
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QUESTION FIFTEEN (Continued)

- (b) Given that
- p
- and
- q
- are constants, use integration by substitution to show that
- 1

$$\int \frac{p-x}{\sqrt{q^2 + (p-x)^2}} dx = -\sqrt{q^2 + (p-x)^2} + c, \text{ where } c \text{ is a constant.}$$

- (c) The diagram below shows an object of mass 1 kg being suspended from the ceiling by two cables of equal length, which are attached to motors at the points
- A
- and
- B
- .



The motors at A and B both pull on their respective cables, creating a tension of T newtons in each cable. The motors are designed such that the magnitude of the tension in each cable is a constant T newtons at all times. The distance between A and B is $2k$ metres, and the ceiling is $2k$ metres above the ground.

Let the height of the object above the ground be x metres, the angle between each cable and the ceiling be θ , and the magnitude of acceleration due to gravity be 10 ms^{-2} .

- (i) Show that
- $\ddot{x} = \frac{2T(2k-x)}{\sqrt{k^2 + (2k-x)^2}} - 10$
- .
- 2

The object is initially at rest on the ground.

- (ii) Using part (b), or otherwise, show that
- $(\dot{x})^2 = 4T \left(k\sqrt{5} - \sqrt{k^2 + (2k-x)^2} \right) - 20x$
- .
- 2

- (iii) Show that if the object touches the ceiling, the minimum magnitude of the force produced by each motor is
- $T = 5\varphi$
- newtons, where
- $\varphi = \frac{1+\sqrt{5}}{2}$
- .
- 1

- (iv) When the value of
- T
- is such that the object just touches the ceiling, the maximum velocity of the object is reached when it has travelled a fraction
- P
- of the distance between the floor and the ceiling.
- 3

Given that $\varphi^2 - 1 = \varphi$, show that $P = \frac{2\sqrt{\varphi} - 1}{2\sqrt{\varphi}}$.

The paper continues on the next page

QUESTION SIXTEEN (15 marks) Start a new answer booklet.

- (a) (i) Let
- $f(x)$
- be a function that is differentiable on the domain
- $x \in [\frac{1}{2}, 1]$
- .
- 2

Use integration by parts to show that

$$\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} f(\sin x) dx = \frac{\pi}{6} (3f(1) - f(\frac{1}{2})) - \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} x \cos x f'(\sin x) dx.$$

- (ii) Hence find
- $\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \frac{2 \sin x + x \cos x}{\sqrt{\sin x}} dx$
- .
- 2

- (b) The AM-GM inequality states that for positive numbers
- $x_1, x_2, \dots, x_n$
- ,

$$\frac{x_1 + x_2 + \dots + x_n}{n} \geq \sqrt[n]{x_1 x_2 \dots x_n}. \quad \text{Do not prove this.}$$

- (i) Show that if
- $a_1, a_2, \dots, a_n$
- are positive numbers, and
- $k_1, k_2, \dots, k_n$
- are positive integers, then
- 1

$$\frac{a_1 k_1 + a_2 k_2 + \dots + a_n k_n}{k_1 + k_2 + \dots + k_n} \geq (a_1^{k_1} a_2^{k_2} \dots a_n^{k_n})^{\frac{1}{k_1 + k_2 + \dots + k_n}}.$$

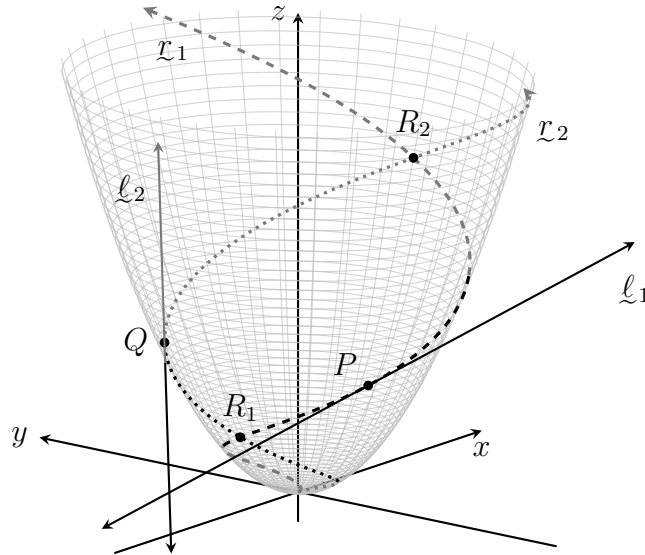
- (ii) Given that the inequality in part (i) is also true when
- $k_1, k_2, \dots, k_n$
- are positive numbers, show that if
- $a, b, c > 0$
- and
- $a + b + c = 1$
- , then
- 3

$$a^a b^b c^c + a^b b^c c^a + a^c b^a c^b \leq 1.$$

The question continues on the next page

QUESTION SIXTEEN (Continued)

- (c) The diagram below shows the paraboloid $x^2 + y^2 = kz$ for some constant $k > 0$, which is the surface formed when the parabola $z = \frac{1}{k}x^2$ is rotated about the z -axis.



The curve r_1 , shown by the dashed curve above, spirals around the paraboloid in an anti-clockwise direction when viewed from above. The curve r_2 , shown by the dotted curve above, spirals around the paraboloid in a clockwise direction.

The point P on the curve r_1 and the point Q on the curve r_2 can be parametrised by

$$P \left(p \cos p, p \sin p, \frac{1}{k}p^2 \right) \quad \text{and} \quad Q \left(q \cos q, -q \sin q, \frac{1}{k}q^2 \right),$$

where p and q are non-negative parameters.

Let ℓ_1 be the tangent to r_1 at P and ℓ_2 be the tangent to r_2 at Q with equations

$$\ell_1 = \begin{pmatrix} p \cos p \\ p \sin p \\ \frac{1}{k}p^2 \end{pmatrix} + \lambda \begin{pmatrix} -p \sin p + \cos p \\ p \cos p + \sin p \\ \frac{2}{k}p \end{pmatrix} \quad \text{and} \quad \ell_2 = \begin{pmatrix} q \cos q \\ -q \sin q \\ \frac{1}{k}q^2 \end{pmatrix} + \mu \begin{pmatrix} -q \sin q + \cos q \\ -q \cos q - \sin q \\ \frac{2}{k}q \end{pmatrix}.$$

The point R_n is the n th point of intersection of the curves r_1 and r_2 above the xy -plane. The points R_1 and R_2 are shown in the diagram above.

- (i) Find, in terms of n , the value of the parameter p that corresponds to the point R_n . 1
- (ii) For all integers $n \geq 1$, $\overrightarrow{R_n R_{n+1}} \perp \overrightarrow{R_{n+1} R_{n+2}}$. 3
- (a) Find the exact value of k . 3
- (b) Let θ_n be the obtuse angle between the two tangents ℓ_1 and ℓ_2 at R_n . 3

Find $\lim_{n \rightarrow \infty} \tan\left(\frac{\theta_n}{2}\right)$.

————— END OF PAPER —————

Maths Ext. 2 Trial 2026

Multiple Choice

① B

② Contrapositive: "If someone misses an exit, then they are not bad at driving". $\Rightarrow$ ①

③ $\int \frac{dx}{\sqrt{5-4x-x^2}} = \int \frac{dx}{\sqrt{9-(x+2)^2}}$

$\Rightarrow$ ⑥

④ $\frac{1}{z} = \frac{1}{x+iy} \times \frac{x-iy}{x-iy}$
 $= \frac{x-iy}{x^2+y^2}$

If $\operatorname{Re}\left(\frac{1}{z}\right) = \frac{1}{2}$, $\frac{x}{x^2+y^2} = \frac{1}{2}$

$$x^2 - 2x + y^2 = 0$$

$$(x-1)^2 + y^2 = 1 \quad \Rightarrow \text{①}$$

⑤ $\operatorname{proj}_{\vec{p}} \vec{q} = \frac{\vec{p} \cdot \vec{q}}{\vec{p} \cdot \vec{p}} \vec{p}$
 $= \frac{\vec{p} \cdot \vec{q}}{|\vec{p}|^2} \times |\vec{p}| \times \hat{\vec{p}}$
 $= \frac{\vec{p} \cdot \vec{q}}{|\vec{p}|} \hat{\vec{p}}$

$$= \frac{1+1+1}{\sqrt{\sin^2\theta + \cos^2\theta + \tan^2\theta}} \hat{\vec{p}}$$

$$= \frac{3}{\sqrt{1+\tan^2\theta}} \hat{\vec{p}}$$

$$= \frac{3}{\sqrt{\sec^2\theta}} \hat{\vec{p}}$$

$$= 3|\cos\theta| \hat{\vec{p}} \quad \therefore \text{Need } |\cos\theta| = \frac{1}{3}, \cos\theta = \pm \frac{1}{3}$$

$\Rightarrow$ 4 solutions ①

$$\textcircled{6} \quad \frac{ir^2}{z} = \frac{ir^2}{re^{i\theta}}$$

$$= ire^{-i\theta}$$

* Conjugate of z , multiplied by i
 $\Rightarrow \textcircled{A}$

$$\textcircled{7} \quad \textcircled{D} \quad \text{i.e.} \quad \exists m, n \in \mathbb{Z}^+ : \sqrt{2+4} = \sqrt{m} + \sqrt{n}$$

$$\sqrt{6} = \sqrt{m} + \sqrt{n}$$

$$6 = m + n + 2\sqrt{mn}$$

Impossible for $m, n \in \mathbb{Z}^+$.

$$\textcircled{8} \quad (\underline{i} - \underline{j} + 2\underline{k}) \cdot (a\underline{i} + b\underline{j} + c\underline{k}) = a - b + 2c$$

$$\therefore a - b + 2c = 0$$

$$A: ((a+c)\underline{i} + (b+c)\underline{j} + c\underline{k}) \cdot (\underline{i} - \underline{j} + 2\underline{k}) = a+c - (b+c) + 2c$$

$$= a - b + 2c$$

$$= 0$$

$$B: (b\underline{i} + a\underline{j} - c\underline{k}) \cdot (\underline{i} - \underline{j} + 2\underline{k}) = b - a - 2c$$

$$= -(a - b + 2c)$$

$$= 0$$

$$C: (b\underline{i} + a\underline{j} + (a-b+c)\underline{k}) \cdot (\underline{i} - \underline{j} + 2\underline{k})$$

$$= b - a + 2a - 2b + 2c$$

$$= a - b + 2c$$

$$= 0$$

$$D: ((2c-a)\underline{i} + (b-2c)\underline{j} + (a+b-c)\underline{k}) \cdot (\underline{i} - \underline{j} + 2\underline{k})$$

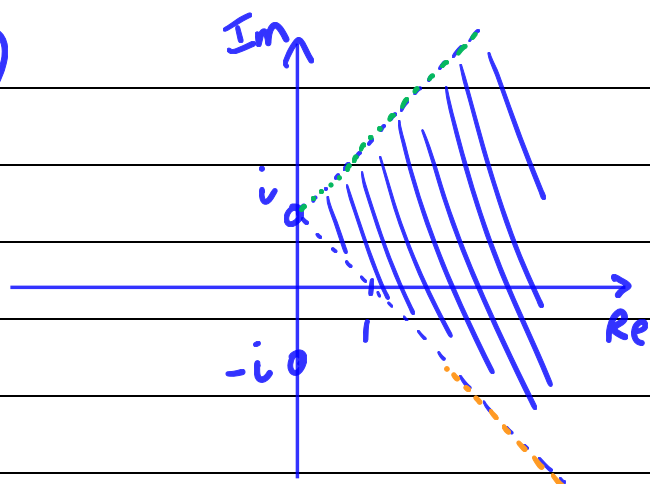
$$= 2c - a + 2c - b + 2a + 2b - 2c$$

$$= a + b + 2c$$

$$\neq 0$$

$\Rightarrow \textcircled{D}$

9



$$\arg(z^2 + 1) = \arg((z+i)(z-i)) \\ = \arg(z+i) + \arg(z-i)$$

When $\arg(z-i) = \frac{\pi}{4}$
 (green branch), $\arg(z+i) < \frac{\pi}{2}$
 $\therefore \arg(z+i) + \arg(z-i) < \frac{3\pi}{4}$

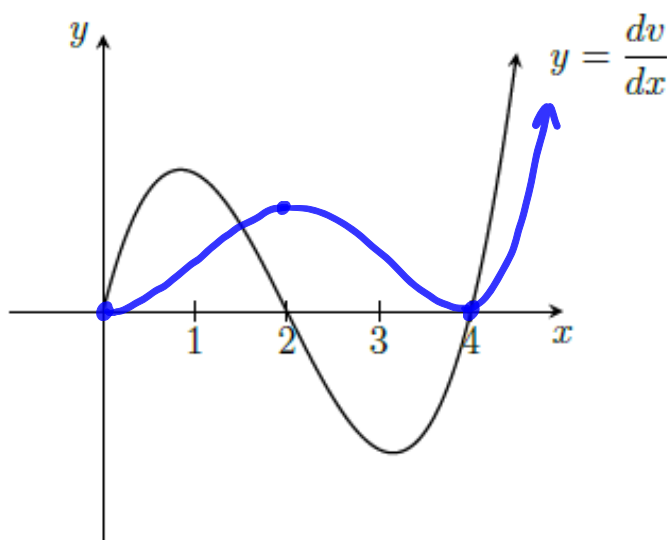
Also, when $\arg(z-i) = -\frac{\pi}{4}$ and $\arg(z+i) < 0$
 (orange branch), $\arg(z+i) > -\frac{\pi}{4}$

$\therefore \arg(z+i) + \arg(z-i) > -\frac{\pi}{2}$
 i.e. $-\frac{\pi}{2} < \arg(z^2 + 1) < \frac{3\pi}{4}$

$\Rightarrow$ (B)

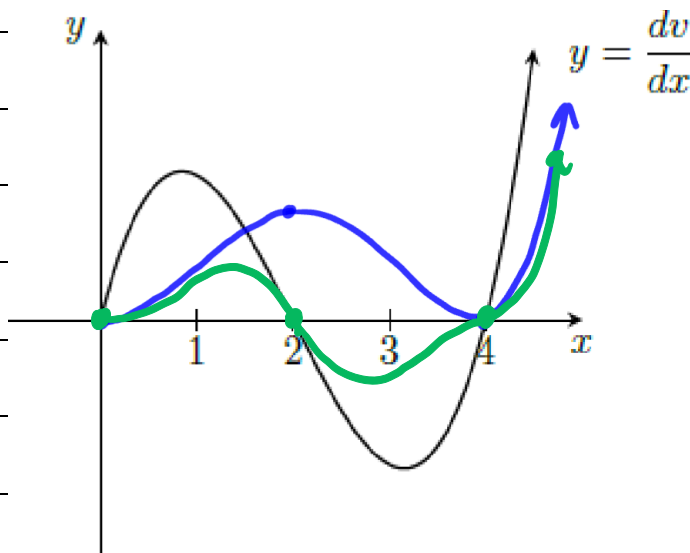
10 $a = v \cdot \frac{dv}{dx}$

Taking the primitive of $\frac{dv}{dx}$
 with $x=0, v=0$ gives:



Taking the product of
 v and $\frac{dv}{dx}$ gives:

$\Rightarrow$ (A)



Question 11

$$(a) \quad z^2 - 4z + 8 = 0$$

$$(z - 2)^2 = -8 + 4$$

$$= -4$$

$$= (2i)^2 \quad \checkmark$$

$$z = 2 \pm 2i \quad \checkmark$$

$$(b) \quad \frac{w}{z} = \frac{8e^{3i\pi/4}}{2e^{i\pi/6}} \\ = 4e^{7i\pi/12}$$

Modulus: 4 $\checkmark$

Argument: $\frac{7\pi}{12}$ $\checkmark$

$$(c) (i) \quad \int (\tan^2 x + 1) \sec^2 x \, dx$$

$$= \int (u^2 + 1) \, du$$

$$= \frac{1}{3} u^3 + u + c$$

$$= \frac{1}{3} \tan^3 x + \tan x + c \quad \checkmark$$

$$u = \tan x$$

$$du = \sec^2 x \, dx$$

$\checkmark$ (recognise substitution)

$$(ii) \quad \int \frac{x^2}{4+x^2} \, dx = \int \left(\frac{x^2+4}{4+x^2} - \frac{4}{4+x^2} \right) \, dx$$

$$= \int \left(1 - \frac{4}{4+x^2} \right) \, dx$$

$$= x - 2 \tan^{-1}\left(\frac{x}{2}\right) + c \quad \checkmark$$

$\checkmark$ (useful algebraic manipulation)

$$(iii) \quad \int x \cos x \, dx = x \sin x - \int \sin x \, dx \quad \checkmark$$

$$v' = \cos x$$

$$u = x$$

$$= x \sin x + \cos x + c \quad \checkmark$$

$$v = \sin x$$

$$u' = 1$$

(d)(i) Solving simultaneously: $\begin{pmatrix} 1+\lambda \\ -2-\lambda \\ 14+3\lambda \end{pmatrix} = \begin{pmatrix} -6+2\mu \\ 3-\mu \\ -1+3\mu \end{pmatrix}$

Equating i : $1+\lambda = -6+2\mu$

$\lambda - 2\mu = -7$ ①

Equating j : $-2-\lambda = 3-\mu$

$\mu - \lambda = 5$ ②

✓ Forming 2 equations

① + ②:

$-\mu = -2$

$\mu = 2$

$\therefore 2 - \lambda = 5$

$\lambda = -3$

Check k :

LHS = $14 + 3\lambda$

$= 14 + 3 \times -3$

$= 5$

RHS = $-1 + 3 \times 2$

$= 5$

✓

$\therefore$ the lines intersect.

When $\mu = 2$, $\vec{r}_2 = \begin{pmatrix} -6+4 \\ 3-2 \\ -1+6 \end{pmatrix}$
 $= \begin{pmatrix} -2 \\ 1 \\ 5 \end{pmatrix}$

$\therefore$ the lines intersect at $(-2, 1, 5)$. ✓

(ii) $\begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} = 2 + 1 + 9$
 $= 12$

$\therefore \cos \theta = \frac{12}{\sqrt{1+1+9} \times \sqrt{4+1+9}}$ ✓

$\theta = \cos^{-1}\left(\frac{12}{\sqrt{11} \times \sqrt{14}}\right)$

$\doteq 15^\circ$ (nearest degree) ✓

Question 12

(a)(i) $n^2 + 7n + 6 = (n+6)(n+1)$

If n is even, then $n+6$ is even, so $n^2 + 7n + 6$ is even. ✓

If n is odd, then $n+1$ is even, so $n^2 + 7n + 6$ is even. ✓

In either case, $n^2 + 7n + 6$ is even.

(ii) Since $n^2 + 7n + 6$ is even, $(n^2 + 7n + 6)^7$ is even.

Also, $16 \times n^5$ is even.

$\therefore (n^2 + 7n + 6)^7 - 16n^5$ is even, but 3 is odd. ✓

$\therefore$ there are 0 integer solutions. ✓

(b)(i) $\frac{3x^2 - 6x + 13}{(x-5)(x^2+4)} = \frac{A}{x-5} + \frac{Bx+C}{x^2+4}$

$$3x^2 - 6x + 13 = A(x^2 + 4) + (Bx + C)(x - 5)$$

$x = 5$: $58 = 29A$

$$A = 2$$

$x = 0$: $13 = 4A - 5C$

$$13 = 8 - 5C$$

$$C = -1$$

✓ (any one correct values)

Equate coef. of x^2 : $A + B = 3$

(ii) $\int \frac{3x^2 - 6x + 13}{(x-5)(x^2+4)} dx = \int \left(\frac{2}{x-5} + \frac{x-1}{x^2+4} \right) dx$ ✓ (all correct values)

$$= \int \left(\frac{2}{x-5} + \frac{x}{x^2+4} - \frac{1}{x^2+4} \right) dx$$

$$= 2 \ln |x-5| + \frac{1}{2} \ln(x^2+4) - \frac{1}{2} \tan^{-1}\left(\frac{x}{2}\right) + C$$

✓ (all correct)

(c) (i) $v^2 = 27 - 18x - 9x^2$

When $v = 0$: $9x^2 + 18x - 27 = 0$

$$x^2 + 2x - 3 = 0$$

$$(x+3)(x-1) = 0$$

$$x = -3 \text{ or } x = 1$$

$\therefore$ Centre of motion: $x = -\frac{-3+1}{2}$

$$= -1$$

amplitude = 2 metres.

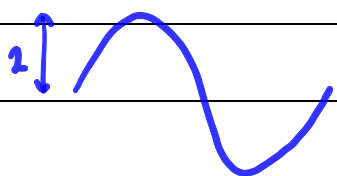
✓ both

(ii) $v^2 = -9(x^2 + 2x - 3)$
 $= -9((x+1)^2 - 4)$

$$\Rightarrow n^2 = 9, \quad n = 3$$

$\therefore$ Period = $\frac{2\pi}{3}$

In 30π seconds, particle completes $\frac{30\pi}{2\pi/3} = 45$ full oscillations.



particle travels 8m/oscillation

$\therefore$ distance travelled: $8 \times 45 = 360$ metres

(iii) $\ddot{x} = -n^2(x-c)$
 $= -9(x+1)$

Max. force acts when furthest from centre of motion.

When $x = -3$, $\ddot{x} = -9x - 2$

$$= 18 \text{ ms}^{-2}$$

$\therefore$ Max force = 5×18
 $= 90 \text{ N}$

(d) (i) As the coefficients are real, $b-ic$ is also a root.

Let the roots be $b-ic, b+ic, \alpha$

$$(b-ic) + (b+ic) + \alpha = -b$$

$$\alpha = -3b$$

$$(ii) \sum \alpha\beta: -3b(b-ic) - 3b(b+ic) + (b-ic)(b+ic) = c$$

$$-3b^2 + 3ibc - 3b^2 - 3ibc + b^2 + c^2 = c$$

$$c^2 - 5b^2 = c$$

①

$$\sum \alpha\beta\gamma: -3b(b-ic)(b+ic) = -3b$$

$$b^2 + c^2 = 1$$

$$b^2 = 1 - c^2$$

②

sub. ② into ①:

$$c^2 - 5(1 - c^2) = c$$

$$6c^2 - c - 5 = 0$$

$$(c-1)(6c+5) = 0$$

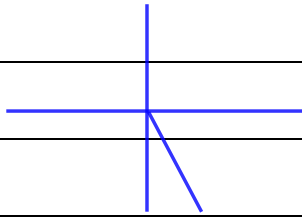
$$c = 1 \text{ or } c = -\frac{5}{6}$$

If $c = 1, b = 0$

$$\text{If } c = -\frac{5}{6}, b^2 = \frac{11}{36}, \text{ so } b = \pm \frac{\sqrt{11}}{6}$$

Question 13

(a) (i)



$$1 - i\sqrt{3} = 2\text{cis}\left(-\frac{\pi}{3}\right)$$

$$\therefore (1 - i\sqrt{3})^n = (2\text{cis}\left(-\frac{\pi}{3}\right))^n \\ = 2^n \text{cis}\left(-\frac{n\pi}{3}\right) \quad \checkmark$$

If n is a multiple of 6, $\exists m \in \mathbb{Z}$ such that $n = 6m$.

$$2^n \text{cis}\left(-\frac{n\pi}{3}\right) = 2^{6m} \text{cis}\left(-\frac{6m\pi}{3}\right) \\ = 2^{6m} \text{cis}(-2m\pi) \\ = 2^{6m} \in \mathbb{R}. \quad \checkmark$$

(ii) Converse: "If $(1 - i\sqrt{3})^n$ is real, then n is a multiple of 6." $\checkmark$ (correctly identify converse)

$(1 - i\sqrt{3})^n = 2^n \text{cis}\left(-\frac{n\pi}{3}\right)$, which is real if $-\frac{n\pi}{3}$ is an integer multiple of π .

This is true when n is a multiple of 3, $\checkmark$ (+ conclusion) so there are values of n (3, 9, 15, ...) where n is not a multiple of 6. $\therefore$ Converse is not true.

(b) $\int e^x \cos 2x \, dx = e^x \cos 2x + 2 \int e^x \sin 2x \, dx \quad \checkmark$

$$v' = e^x \quad u = \cos 2x$$

$$v = e^x \quad u' = -2\sin 2x$$

$$v' = e^x \quad u = \sin 2x$$

$$v = e^x \quad u' = 2\cos 2x$$

$$= e^x \cos 2x + 2 \left(e^x \sin 2x - 2 \int e^x \cos 2x \, dx \right) \quad \checkmark$$

$$\therefore I = e^x \cos 2x + 2e^x \sin 2x - 4I$$

$$5I = e^x \cos 2x + 2e^x \sin 2x + c_1$$

$$I = \frac{1}{5} (e^x \cos 2x + 2e^x \sin 2x) + c_2 \quad \checkmark$$

$$(c) (i) |\underline{a}| = |\underline{b} - \underline{a}|$$

$$\therefore \underline{a} \cdot \underline{a} = (\underline{b} - \underline{a}) \cdot (\underline{b} - \underline{a})$$

$$\cancel{\underline{a} \cdot \underline{a}} = \underline{b} \cdot \underline{b} - 2\underline{a} \cdot \underline{b} + \cancel{\underline{a} \cdot \underline{a}}$$

$$\therefore \underline{b} \cdot \underline{b} = 2\underline{a} \cdot \underline{b} \quad \checkmark$$

$$(ii) \text{ From (i): } \underline{b} \cdot \underline{b} = 2\underline{a} \cdot \underline{b}$$

$$\text{Similarly, } |\underline{c}| = |\underline{b} - \underline{c}| \text{ gives } \underline{b} \cdot \underline{b} = 2\underline{b} \cdot \underline{c} \quad \checkmark$$

$$\vec{OB} \cdot \vec{AC} = \underline{b} \cdot (\underline{c} - \underline{a})$$

$$= \underline{b} \cdot \underline{c} - \underline{b} \cdot \underline{a}$$

$$= \frac{1}{2} \underline{b} \cdot \underline{b} - \frac{1}{2} \underline{b} \cdot \underline{b} \quad \text{from above}$$

$$= 0 \quad \checkmark$$

$$\therefore \vec{OB} \perp \vec{AC}$$

$$(d) (i) (\cos \theta + i \sin \theta)^4$$

$$= \cos^4 \theta + 4\cos^3 \theta (i \sin \theta) + 6\cos^2 \theta (i \sin \theta)^2 + 4\cos \theta (i \sin \theta)^3 + (i \sin \theta)^4 \quad \checkmark \text{ (correct expansion)}$$

$$= \underbrace{\cos^4 \theta - 6\cos^2 \theta \sin^2 \theta + \sin^4 \theta}_{\cos^4 \theta} + i \underbrace{(4\cos^3 \theta \sin \theta - 4\cos \theta \sin^3 \theta)}_{\sin^4 \theta \text{ by De Moivre's}} \quad \checkmark$$

$$\tan 4\theta = \frac{4\cos^3 \theta \sin \theta - 4\cos \theta \sin^3 \theta}{\cos^4 \theta - 6\cos^2 \theta \sin^2 \theta + \sin^4 \theta} = \frac{\cos^4 \theta}{\cos^4 \theta} \quad \checkmark$$

$$= \frac{4\tan \theta - 4\tan^3 \theta}{1 - 6\tan^2 \theta + \tan^4 \theta}$$

Must show some understanding of how to reach last line

$$(ii) \text{ Let } \tan 4\theta = -1: \quad \checkmark$$

$$-1 = \frac{4\tan \theta - 4\tan^3 \theta}{1 - 6\tan^2 \theta + \tan^4 \theta}$$

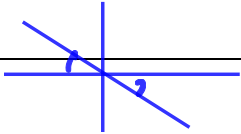
$$-1 + 6\tan^2\theta - \tan^4\theta = 4\tan\theta - 4\tan^3\theta$$

$$\tan^4\theta - 4\tan^3\theta - 6\tan^2\theta + 4\tan\theta + 1 = 0$$

Letting $x = \tan\theta$:

$$x^4 - 4x^3 - 6x^2 + 4x + 1 = 0$$

$\therefore$ the roots of the polynomial are $x = \tan\theta$,
where $\tan 4\theta = -1$.



$$\tan 4\theta = -1$$

$$4\theta = \pi - \frac{\pi}{4}, 2\pi - \frac{\pi}{4}, 3\pi - \frac{\pi}{4}, 4\pi - \frac{\pi}{4}$$

$$= \frac{3\pi}{4}, \frac{7\pi}{4}, \frac{11\pi}{4}, \frac{15\pi}{4}$$

$$\theta = \frac{3\pi}{16}, \frac{7\pi}{16}, \frac{11\pi}{16}, \frac{15\pi}{16}$$

$\therefore$ roots are $\tan \frac{3\pi}{16}, \tan \frac{7\pi}{16}, \tan \frac{11\pi}{16}, \tan \frac{15\pi}{16}$. ✓

(Accept θ in any domain
such that all $\tan\theta$ are
unique)

Question 14

(a) (i) $\ddot{x} = -g + kv^2$

When $v = 40$, $\ddot{x} = 0$:

$$0 = -g + k \times (-40)^2$$

$$\frac{g}{k} = 40^2$$

$$\sqrt{\frac{g}{k}} = 40 \quad \checkmark$$

(ii) (α) $\ddot{x} = -g - kv^2$

$$\frac{dv}{dt} = -(g + kv^2)$$

$$\int \frac{dv}{g + kv^2} = -\int dt$$

$$\frac{1}{k} \int \frac{dv}{\frac{g}{k} + v^2} = -\int dt$$

$$\int \frac{dv}{40^2 + v^2} = -k \int dt$$

$$\frac{1}{40} \tan^{-1}\left(\frac{v}{40}\right) = -kt + c \quad \checkmark$$

When $t = 0$, $v = 40$: $c = \frac{1}{40} \times \tan^{-1}(1)$
 $= \frac{\pi}{160}$

$$\tan^{-1}\left(\frac{v}{40}\right) = 40\left(\frac{\pi}{160} - kt\right)$$

$$\frac{v}{40} = \tan\left(\frac{\pi}{4} - 40kt\right)$$

$$v = 40 \tan\left(\frac{\pi}{4} - 40kt\right) \quad \checkmark$$

(β) When $t = 4$, $v = 0$

$$\therefore \tan\left(\frac{\pi}{4} - 160k\right) = 0$$

$$\frac{\pi}{4} - 160k = 0$$

$$k = \frac{\pi}{640} \quad \checkmark$$

$$\frac{g}{k} = 40^2$$

$$g = 40^2 \times k = 40^2 \times \frac{\pi}{640} = \frac{5\pi}{2} \text{ ms}^{-2} \quad \checkmark$$

$$\begin{aligned}
 (b) \quad (i) \quad e^{i\theta} + e^{-i\theta} &= \cos\theta + i\sin\theta + \cos(-\theta) + i\sin(-\theta) \\
 &= \cos\theta + i\sin\theta + \cos\theta - i\sin\theta \quad (\cos \text{ even, } \sin \text{ odd}) \\
 &= 2\cos\theta
 \end{aligned}$$

$$(ii) \quad \text{Since } |z_3| = \frac{1}{\sqrt{3}},$$

$$\left| \frac{1}{z_1 + z_2} \right| = \frac{1}{\sqrt{3}}$$

$$\therefore |z_1 + z_2| = \sqrt{3}$$

$$|e^{i\alpha} + e^{i\beta}| = \sqrt{3}$$

$$\therefore \left| e^{i(\alpha+\beta)/2} (e^{i(\alpha-\beta)/2} + e^{-i(\alpha-\beta)/2}) \right| = \sqrt{3}$$

$$|e^{i(\alpha+\beta)/2}| \times |e^{i(\alpha-\beta)/2} + e^{-i(\alpha-\beta)/2}| = \sqrt{3}$$

$$1 \times |2\cos\left(\frac{\alpha-\beta}{2}\right)| = \sqrt{3} \quad (\text{from (i)})$$

$$|\cos\left(\frac{\alpha-\beta}{2}\right)| = \frac{\sqrt{3}}{2}$$

$$\cos\frac{\alpha-\beta}{2} = \pm \frac{\sqrt{3}}{2}$$

$$\text{Since } 0 \leq \alpha, \beta \leq \pi, \quad -\frac{\pi}{2} \leq \frac{\alpha-\beta}{2} \leq \frac{\pi}{2}$$

$$\cos\left(\frac{\alpha-\beta}{2}\right) = \frac{\sqrt{3}}{2}$$

$$\cos\left(\frac{\alpha-\beta}{2}\right) = -\frac{\sqrt{3}}{2}$$

$$\Rightarrow \frac{\alpha-\beta}{2} = \frac{\pi}{6}$$

$$\Rightarrow \frac{\alpha-\beta}{2} = -\frac{\pi}{6}$$

$$\alpha - \beta = \frac{\pi}{3}$$

$$\alpha - \beta = -\frac{\pi}{3}$$

✓ (both values)

(c) (i) Base case: Prove true for $n=1$.

$$\begin{aligned}x^n + 1 &= x + 1 \\&= 1(x+1)\end{aligned}$$

$\therefore$ true for $n=1$

Suppose statement is true for some $n=k$.

$$\text{i.e. } x^k + 1 = M(x+1), \text{ where } M \in \mathbb{Z}. \quad (*)$$

Prove true for $n=k+2$:

RTP: $x^{k+2} + 1$ is divisible by $x+1$.

$$\begin{aligned}x^{k+2} + 1 &= x^2 \times x^k + 1 \\&= x^2(M(x+1) - 1) + 1 \quad \text{from } (*) \quad \checkmark \\&= x^2 M(x+1) - x^2 + 1 \\&= Mx^2(x+1) - (x^2 - 1) \\&= Mx^2(x+1) - (x+1)(x-1) \\&= (x+1)(Mx^2 - (x-1)) \quad \checkmark\end{aligned}$$

$\therefore$ true for $n=k+2$.

$\therefore$ statement is true for all positive odd integers
by mathematical induction.

(ii) $n = m \times 2^k$

Contrapositive: If m does not have an odd factor
larger than 1, then n is a power of 2. $\checkmark$

If m doesn't have an odd factor > 1 , then m
is a power of 2. $\therefore$ Let $m = 2^j$, $j \in \mathbb{Z}^+$.

$$\begin{aligned}\therefore n &= 2^j \times 2^k \\&= 2^{j+k}, \quad \checkmark \text{ which is a power of 2.}\end{aligned}$$

$\therefore$ By the contrapositive, if n is not a power of 2, then
 m has an odd factor larger than 1.

(iii) Again, consider the contrapositive:

"If n is not a power of 2, then $2^n + 1$ is not prime". ✓

From part (ii), if n is not a power of 2, then

$n = m \times 2^k$, where m has an odd factor > 1 .

Let $n = a \times b$, where a is odd and $a > 1$,
and $b > 1$.

$$\begin{aligned}\text{Then } 2^n + 1 &= 2^{ab} + 1 \\ &= (2^b)^a + 1\end{aligned}$$

From part (i), this is divisible by $2^b + 1$

since a is odd. Also, since $b > 1$, $2^b + 1 > 1$

$\therefore (2^b)^a + 1 = 2^n + 1$ is not prime. ✓

By contraposition, if $2^n + 1$ is prime, then n is
a power of 2.

Question 15

$$(a) \quad I_n = \int_0^1 (1-\sqrt{x})^n dx$$

$$(i) \quad \int_0^1 (1-\sqrt{x})^n dx = \int_0^1 1 \times (1-\sqrt{x})^n dx$$

$$v' = 1 \quad u = (1-\sqrt{x})^n$$

$$v = x \quad u' = n(1-\sqrt{x})^{n-1} \times -\frac{1}{2}x^{-\frac{1}{2}}$$

$$= \left[x(1-\sqrt{x})^n \right]_0^1 + \lim_{n \rightarrow 0} \frac{n}{2} \int_0^1 x(1-\sqrt{x})^{n-1} \times x^{-\frac{1}{2}} dx$$

$$= \frac{n}{2} \int_0^1 \sqrt{x} (1-\sqrt{x})^{n-1} dx$$

$$= \frac{n}{2} \int_0^1 -((1-\sqrt{x})-1)(1-\sqrt{x})^{n-1} dx$$

$$= -\frac{n}{2} \int_0^1 \left((1-\sqrt{x})^n - (1-\sqrt{x})^{n-1} \right) dx$$

$$I_n = -\frac{n}{2} (I_n - I_{n-1})$$

$$I_n \left(1 + \frac{n}{2} \right) = \frac{n}{2} I_{n-1}$$

$$I_n \left(\frac{n+2}{2} \right) = \frac{n}{2} I_{n-1}$$

$$I_n = \frac{n}{n+2} I_{n-1}$$

$$(ii) \quad I_n = \frac{n}{n+2} I_{n-1}$$

$$= \frac{n}{n+2} \times \frac{n-1}{n+1} \times I_{n-2}$$

$$= \frac{n}{n+2} \times \frac{n-1}{n+1} \times \frac{n-2}{n} \times I_{n-3}$$

$$= \frac{\cancel{n}}{n+2} \times \frac{\cancel{n-1}}{n+1} \times \frac{\cancel{n-2}}{\cancel{n}} \times \frac{\cancel{n-3}}{\cancel{n-1}} \times \dots \times \frac{\cancel{3}}{\cancel{5}} \times \frac{2}{4} \times \frac{1}{3} \times I_0$$

$$= \frac{1}{n+2} \times \frac{1}{n+1} \times 2 \times 1 \quad (\text{since } I_0 = 1)$$

$$= \frac{2}{(n+2)(n+1)} \quad \checkmark \quad (\text{must address } I_0)$$

$$(iii) \text{ Let } \frac{2}{(n+1)(n+2)} = \frac{A}{n+1} + \frac{B}{n+2}$$

$$2 = A(n+2) + B(n+1)$$

$$n=-1: A=2$$

$$n=-2: B=-2$$

$$\therefore \frac{2}{(n+1)(n+2)} = \frac{2}{n+1} - \frac{2}{n+2} \quad \checkmark$$

$$I_0 + I_1 + I_2 + I_3 + \dots + I_{n-1} + I_n$$

$$= \left(\frac{2}{1} - \frac{2}{2} \right) + \left(\frac{2}{2} - \frac{2}{3} \right) + \left(\frac{2}{3} - \frac{2}{4} \right) + \dots + \left(\frac{2}{n} - \frac{2}{n+1} \right) + \left(\frac{2}{n+1} - \frac{2}{n+2} \right)$$

$$= 2 - \frac{2}{n+2}$$

$$< 2 \quad \text{since } n > 0 \quad \checkmark$$

$$(b) \int \frac{p-x}{\sqrt{q^2+(p-x)^2}} dx \quad \text{Let } u = (p-x)^2$$

$$du = -2(p-x) dx$$

$$= -\frac{1}{2} \int \frac{-2(p-x) dx}{\sqrt{q^2+(p-x)^2}}$$

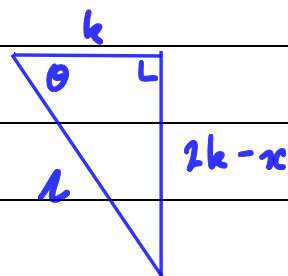
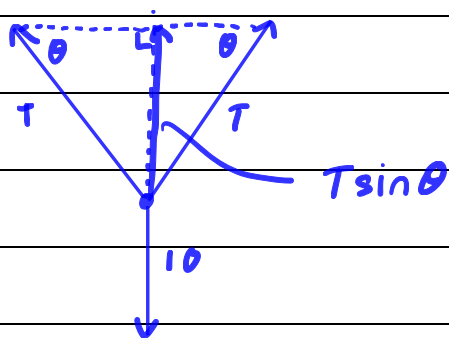
$$= -\frac{1}{2} \int (q^2+u)^{-\frac{1}{2}} du$$

$$= -\frac{1}{2} \times \frac{1}{\frac{1}{2}} (q^2+u)^{\frac{1}{2}} + c$$

$$= -\sqrt{q^2+u} + c$$

$$= -\sqrt{q^2+(p-x)^2} + c \quad \checkmark$$

(c) (i)



$$k = \sqrt{k^2 + (2k - x)^2}$$

$$\sin \theta = \frac{2k - x}{\sqrt{k^2 + (2k - x)^2}} \quad \checkmark$$

Since mass = 1 kg,

$$\ddot{x} = 2T \sin \theta - 10$$

$$= \frac{2T(2k - x)}{\sqrt{k^2 + (2k - x)^2}} - 10 \quad \checkmark$$

(ii) $\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = \frac{2T(2k - x)}{\sqrt{k^2 + (2k - x)^2}} - 10$

$$\frac{1}{2} v^2 = -2T \sqrt{k^2 + (2k - x)^2} - 10x + c \quad \text{from part (a)} \quad \checkmark$$

$$v^2 = -4T \sqrt{k^2 + (2k - x)^2} - 20x + c_2$$

When $x = 0$, $v = 0$: $c_2 = 4T \sqrt{5k^2}$
 $= 4Tk\sqrt{5}$

$$\therefore v^2 = 4T(k\sqrt{5} - \sqrt{k^2 + (2k - x)^2}) - 20x$$

i.e. $(\dot{x})^2 = 4T(k\sqrt{5} - \sqrt{k^2 + (2k - x)^2}) - 20x \quad \checkmark$

(iii) If object touches the ceiling, $\dot{x} > 0$ when $x = 2k$.

i.e. $4T(k\sqrt{5} - \sqrt{k^2}) - 20 \times 2k > 0$

$$T > \frac{40k}{4k(\sqrt{5} - 1)} \times \frac{\sqrt{5} + 1}{\sqrt{5} + 1}$$

$$T > \frac{10(\sqrt{5} + 1)}{4}$$

$$T > 5\varphi, \quad \text{where } \varphi = \frac{\sqrt{5} + 1}{2} \quad \checkmark$$

(iv) Maximum velocity reached when $\ddot{x} = 0$.

Using $T = 5\varphi$:

$$\frac{10\varphi(2k-x)}{\sqrt{k^2+(2k-x)^2}} - 10 = 0 \quad \checkmark$$

$$\varphi(2k-x) = \sqrt{k^2+(2k-x)^2}$$

$$\varphi^2(2k-x)^2 = k^2+(2k-x)^2$$

$$(2k-x)^2(\varphi^2-1) = k^2$$

$$(2k-x)^2 = \frac{k^2}{\varphi^2-1}$$

$$= \frac{k^2}{\varphi} \quad \checkmark$$

$$2k-x = \frac{k}{\sqrt{\varphi}} \quad \text{since } 0 \leq x \leq 2k \text{ and } k > 0$$

$$x = \frac{2k\sqrt{\varphi} - k}{\sqrt{\varphi}}$$

$\therefore$ Fraction of way between floor and ceiling:

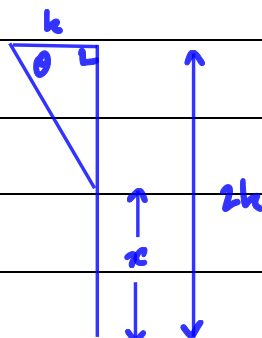
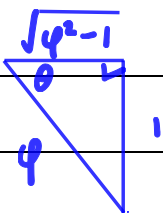
$$p = \frac{\frac{k(2\sqrt{\varphi}-1)}{\sqrt{\varphi}}}{2k}$$

$$= \frac{2\sqrt{\varphi}-1}{2\sqrt{\varphi}} \quad \checkmark$$

Alternatively: $\ddot{x} = 2T \sin \theta - 10$

$$\ddot{x} = 0, \quad T = 5\varphi: \quad 10\varphi \sin \theta - 10 = 0$$

$$\sin \theta = \frac{1}{\varphi}$$



$$\begin{aligned} p &= 1 - \frac{2k-x}{2k} \\ &= 1 - \frac{1}{2} \times \frac{2k-x}{k} \\ &= 1 - \frac{1}{2} \times \tan \theta \\ &= 1 - \frac{1}{2} \times \frac{1}{\sqrt{\varphi^2-1}} \\ &= \frac{2\sqrt{\varphi}-1}{2\sqrt{\varphi}} \end{aligned}$$

Question 16

(a) (i) $\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} f(\sin x) dx$

$$\begin{array}{ll} v' = 1 & u = f(\sin x) \\ v = x & u' = f'(\sin x) \cos x \end{array}$$

$$= \left[x f(\sin x) \right]_{\frac{\pi}{6}}^{\frac{\pi}{2}} - \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} x \cos x f'(\sin x) dx \quad \checkmark$$

$$= \frac{\pi}{2} f(\sin \frac{\pi}{2}) - \frac{\pi}{6} f(\sin \frac{\pi}{6}) - \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} x \cos x f'(\sin x) dx$$

$$= \frac{\pi}{6} \left(3f(1) - f\left(\frac{1}{2}\right) \right) - \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} x \cos x f'(\sin x) dx \quad \checkmark$$

(ii) From (i): $\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \left(f(\sin x) + x \cos x f'(\sin x) \right) dx = \frac{\pi}{6} \left(3f(1) - f\left(\frac{1}{2}\right) \right)$

$$\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \frac{2 \sin x - x \cos x}{\sqrt{\sin x}} dx = \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \left(2\sqrt{\sin x} - \frac{x \cos x}{\sqrt{\sin x}} \right) dx$$

Let $f(x) = 2\sqrt{x}$ ✓

$$f'(x) = x^{-\frac{1}{2}} = \frac{1}{\sqrt{x}}$$

$$\text{So } \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \left(2\sqrt{\sin x} - \frac{x \cos x}{\sqrt{\sin x}} \right) dx = \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \left(f(\sin x) + x \cos x f'(\sin x) \right) dx$$

(where $f(x) = 2\sqrt{x}$)

$$= \frac{\pi}{6} \left(3f(1) - f\left(\frac{1}{2}\right) \right)$$

$$= \frac{\pi}{6} \left(3 \times 2\sqrt{1} - 2\sqrt{\frac{1}{2}} \right)$$

$$= \frac{\pi}{6} (6 - \sqrt{2}) \quad \checkmark$$

(b) (i) Consider the numbers:

$$\underbrace{a_1, a_1, \dots, a_1}_{k_1}, \underbrace{a_2, a_2, \dots, a_2}_{k_2}, \dots, \underbrace{a_n, a_n, \dots, a_n}_{k_n} \quad \checkmark$$

Applying AM-GM:

$$\frac{a_1 + a_1 + \dots + a_1 + a_2 + a_2 + \dots + a_2 + a_n + a_n + \dots + a_n}{k_1 + k_2 + \dots + k_n} \geq \left(a_1 \times a_1 \times \dots \times a_1 \times a_2 \times a_2 \times \dots \times a_2 \times a_n \times a_n \times \dots \times a_n \right)^{\frac{1}{k_1 + k_2 + \dots + k_n}}$$

$$\therefore \frac{a_1 k_1 + a_2 k_2 + \dots + a_n k_n}{k_1 + k_2 + \dots + k_n} \geq (a_1^{k_1} a_2^{k_2} \dots a_n^{k_n})^{\frac{1}{k_1 + k_2 + \dots + k_n}}$$

(ii) Let $a_1 = a, k_1 = a, a_2 = b, k_2 = b, a_3 = c, k_3 = c$:

$$\frac{a^2 + b^2 + c^2}{a + b + c} \geq (a^a b^b c^c)^{\frac{1}{a+b+c}}$$

Since $a+b+c=1$: $a^2 + b^2 + c^2 \geq a^a b^b c^c$ (1) $\checkmark$

Let $a_1 = a, k_1 = b, a_2 = b, k_2 = c, a_3 = c, k_3 = a$:

$$\frac{ab + bc + ac}{a + b + c} \geq (a^b b^c c^a)^{\frac{1}{a+b+c}}$$

Since $a+b+c=1$: $ab + bc + ac \geq a^b b^c c^a$ (2) $\checkmark$

Let $a_1 = a, k_1 = c, a_2 = b, k_2 = a, a_3 = c, k_3 = b$:

$$\frac{ac + ab + bc}{a + b + c} \geq (a^c b^a c^b)^{\frac{1}{a+b+c}}$$

Since $a+b+c=1$: $ac + ab + bc \geq a^c b^a c^b$ (3) $\checkmark$

$$\begin{aligned} \textcircled{1} + \textcircled{2} + \textcircled{3}: \quad a^a b^b c^c + a^b b^c c^a + a^c b^a c^b &\leq a^2 + b^2 + c^2 + 2(ab + ac + bc) \\ &= (a+b+c)^2 \\ &= 1 \quad \text{since } a+b+c=1 \end{aligned} \quad \checkmark$$

(c) (i) Solving simultaneously:
$$\begin{pmatrix} p \cos p \\ p \sin p \\ \frac{1}{k} p^2 \end{pmatrix} = \begin{pmatrix} q \cos q \\ -q \sin q \\ \frac{1}{k} q^2 \end{pmatrix}$$

i and k coordinates equal when $p=q$, but j coordinates will only be equal when $p=q$ and coordinates are 0.

i.e. $p \sin p = 0$, so R_1 is denoted by $p = \pi$,

R_2 by 2π , etc.

$\therefore R_n$ has parameter $n\pi$. ✓

(ii) (x)
$$R_n = \begin{pmatrix} n\pi \cos(n\pi) \\ n\pi \sin(n\pi) \\ \frac{1}{k} (n\pi)^2 \end{pmatrix} = \begin{pmatrix} n\pi \cos(n\pi) \\ 0 \\ \frac{1}{k} (n\pi)^2 \end{pmatrix}$$

$$R_{n+1} = \begin{pmatrix} (n+1)\pi \cos((n+1)\pi) \\ 0 \\ \frac{1}{k} ((n+1)\pi)^2 \end{pmatrix} = \begin{pmatrix} -(n+1)\pi \cos(n\pi) \\ 0 \\ \frac{1}{k} ((n+1)\pi)^2 \end{pmatrix}$$

$$\overrightarrow{R_n R_{n+1}} = \begin{pmatrix} -(n+1)\pi \cos(n\pi) - n\pi \cos(n\pi) \\ 0 \\ \frac{1}{k} ((n+1)\pi)^2 - \frac{1}{k} (n\pi)^2 \end{pmatrix}$$

$$= \begin{pmatrix} -(2n+1)\pi \cos(n\pi) \\ 0 \\ \frac{1}{k} \pi^2 (2n+1) \end{pmatrix} \quad \checkmark$$

$$\therefore \overrightarrow{R_{n+1} R_{n+2}} = \begin{pmatrix} -(2(n+1)+1)\pi \cos((n+1)\pi) \\ 0 \\ \frac{1}{k} \pi^2 (2(n+1)+1) \end{pmatrix}$$

$$= \begin{pmatrix} (2n+3)\pi \cos(n\pi) \\ 0 \\ \frac{1}{k} \pi^2 (2n+3) \end{pmatrix}$$

$$\vec{R_n R_{n+1}} \cdot \vec{R_{n+1} R_{n+2}} = -(2n+1)(2n+3)\pi^2 \cos^2(n\pi) + \frac{\pi^4}{k^2} (2n+1)(2n+3)$$

Since $\vec{R_n R_{n+1}} \perp \vec{R_{n+1} R_{n+2}}$:

$$-(2n+1)(2n+3)\pi^2 \cos^2(n\pi) + \frac{\pi^4}{k^2} (2n+1)(2n+3) = 0$$

As $\cos^2(n\pi) = 1$: $\frac{\pi^2}{k^2} = 1$

* Note: Award full marks if

$$k^2 = \pi^2$$

a pupil uses $\vec{R_1 R_2} \cdot \vec{R_2 R_3} = 0$

$$k = \pi \quad \checkmark \text{ (since } k > 0 \text{)}$$

(i.e. picking a specific n)

(ii)(b) Let the direction vector of $\underline{l_1}$ be $\underline{d_1}$, and $\underline{l_2}$ be $\underline{d_2}$.

At R_n :

$$\underline{d_1} = \begin{pmatrix} -n\pi \sin(n\pi) + \cos(n\pi) \\ n\pi \cos(n\pi) + \sin(n\pi) \\ \frac{2}{\pi} \times n\pi \end{pmatrix}$$

$$= \begin{pmatrix} \cos(n\pi) \\ n\pi \cos(n\pi) \\ 2n \end{pmatrix}$$

$$\underline{d_2} = \begin{pmatrix} -n\pi \sin(n\pi) + \cos(n\pi) \\ -n\pi \cos(n\pi) - \sin(n\pi) \\ \frac{2}{\pi} \times n\pi \end{pmatrix}$$

$$= \begin{pmatrix} \cos(n\pi) \\ -n\pi \cos(n\pi) \\ 2n \end{pmatrix}$$

✓ Simplified form of direction vectors.

$$\cos \theta_n = \frac{\underline{d_1} \cdot \underline{d_2}}{|\underline{d_1}| \times |\underline{d_2}|}$$

$$= \frac{\cos^2(n\pi) - n^2 \pi^2 \cos^2(n\pi) + 4n^2}{(\sqrt{\cos^2(n\pi) + n^2 \pi^2 \cos^2(n\pi) + 4n^2})^2}$$

$$= \frac{1 - n^2 \pi^2 + 4n^2}{1 + n^2 \pi^2 + 4n^2} \quad \checkmark$$

Note that $\cos x = \frac{1 - \tan^2 \frac{x}{2}}{1 + \tan^2 \frac{x}{2}}$

$$\begin{aligned}\cos \theta_n &= \frac{1 + 4n^2 - n^2 \pi^2}{1 + 4n^2 + n^2 \pi^2} \\ &= \frac{1 - \frac{n^2 \pi^2}{1 + 4n^2}}{1 + \frac{n^2 \pi^2}{1 + 4n^2}}\end{aligned}$$

$$\therefore \tan^2 \frac{\theta_n}{2} = \frac{n^2 \pi^2}{1 + 4n^2}$$

$$\tan \frac{\theta_n}{2} = \frac{n\pi}{\sqrt{1+4n^2}} \quad \text{since } \theta_n \text{ is obtuse so } \tan \frac{\theta_n}{2} > 0.$$

$$= \frac{n\pi}{n\sqrt{\frac{1}{n^2} + 4}}$$

$$= \frac{\pi}{\sqrt{\frac{1}{n^2} + 4}}$$

$$\begin{aligned}\therefore \lim_{n \rightarrow \infty} \tan \frac{\theta_n}{2} &= \frac{\pi}{\sqrt{4}} \\ &= \frac{\pi}{2} \quad \checkmark\end{aligned}$$